Remains Anomaly Pipeline

Multi-layer search prioritization for buried human remains: fuses multi-temporal lidar, satellite NDVI, case reports, and soil nitrate into a calibrated P(grave | data) map, with a qPCR confirmation module.

Companion modules

- `run_county.py` - CLI driver (run / validate / demo / curve)
- `field_confirmation.py` - qPCR assay QC (LOD95/LOQ, triplicate concordance, per-sample decision logic) + culvert/void detector (black top-hat + elongation) for the trapped-in-culvert/vault search case

Layers

#	Layer	Input	Output
A	Lidar timeline	USGS 3DEP EPT (PDAL), OpenTopography DEMs, drone DTMs	mound->sink signature raster
B	NDVI anomaly	Sentinel-2 L2A, 3-yr same-season baseline	decomposition-island z-map
C	Report fusion	NamUs / CAD / geoparsed tips	kernel-density prior
D	Nitrate probes	field ion-selective electrode, 10-20 cores/candidate	IDW hotspot (non-specific!)
E	qPCR confirm	soil cores, humic-removing kit, HV1 miniamplicons	copies/g + interpretive call

Plus: slope-stratified Getis-Ord Gi*, LoD-propagated change confidence, spatially blocked CV calibration, LOD experiment, real-site label adapter.

Install

```
conda install -c conda-forge pdal numpy scipy pandas rasterio requests
pip install scikit-learn
```

Quickstart

```
python remains_anomaly_pipeline.py        # full demo + LOD experiment
python run_county.py demo                 # county plumbing integration test
python run_county.py curve                # operating-curve experiment + chart
python run_county.py validate config.json # config check, no data pulled
python run_county.py run config.json      # real county run
```

Operating curve (run_county.py curve)

Site-level PR/F1 sweep of the P(grave) gate. Synthetic result: at 3DEP-grade noise a 6 cm decay sink is sub-LoD and the gate is undecidable; drone-grade (3 cm LoD) supports P>0.15-0.25 at precision 1.0 / recall 0.75. Thresholds are sensor-grade-specific - recalibrate per deployment, never copy them.

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Real data

1. Find a 3DEP EPT project URL in your AOI (USGS / AWS Open Data registry)
2. Build epochs list -> ingest_epochs([...]) # year-stamped DEMs
3. Reports -> report_density(points, shape, res)
4. Probe candidates -> nitrate_hotspot(samples, shape, res)
5. Labels -> labels_from_sites_csv(path, shape, res, origin_xy, years) (sites buried before the first lidar epoch are censored, not negatives)
6. feature_dataframe(rasters, lab, groups=block_groups(shape, 25)) -> calibrate_probability(df, feats, model_type="gbm") # or "logistic"
7. Multi-tile: county_sweep(tiles, model, feats) -> one deduped ranked list (candidates_from_prob reports world coordinates; dedupe_candidates removes edge double-counts)
8. rank by probability_raster(model, rasters, feats), dig, then qPCR confirm

Caveats (read before operational use)

- Calibrated on synthetic Gaussian graves; re-calibrate on real labeled sites.
- AUC from blocked CV is the honest number; random folds leak ~0.15 AUC.
- Nitrate is nonspecific (fertilizer, septic, livestock) - never proof.
- Lidar blind to burials predating the first epoch (use reports/dogs).
- Every positive is "probable cause to dig" - only Layer E confirms.

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